

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Electromagnetism Lab

L-T-P: 0-0-3

Credits: 2

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

1. Coupled Oscillator.
2. Inductance of Solenoid
3. Magnetic moment
4. Transformer
5. Fresnel's Equation
6. e/m by Helmholtz coils
7. e/m by helical method.
8. Self-inductance of a given coil by Anderson's bridge.
9. Series and parallel L-C-R circuit by CRO and function generator.
10. High resistance by the method of leakage of charge of a charged conductor.
11. Self-inductance of a given coil by Owen's bridge.
12. Refraction, interference, diffraction, absorption and polarization of the microwaves.

Reference Books:

Lab manual